Grid Forming Mechanism

Laboratory-Scale , Three-Phase Grid-Forming Inverter for Islanded Microgrid Applications

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1 Control algorithm

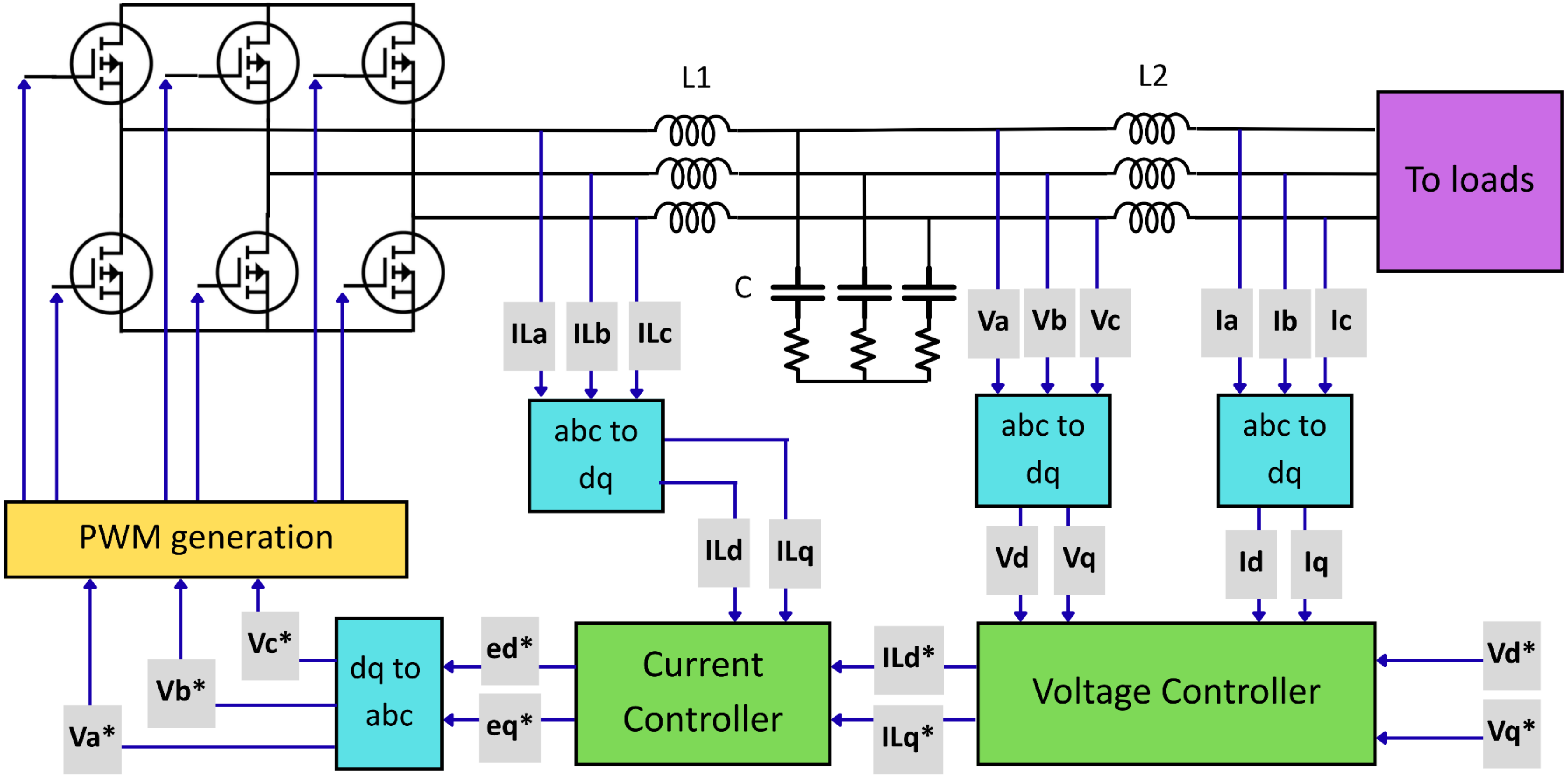

Figure 1: Control algorithm

2 Control algorithm input parameters from the hardware setup

Table 1:Input parameters for control algorithm

Input parameter	
Va	AC voltage across capacitors of LCL filter (phase A)
Vb	AC voltage across capacitors of LCL filter (phase B)
Vc	AC voltage across capacitors of LCL filter (phase C)
Ila	Phase current before LCL filter (phase A)
ILb	Phase current before LCL filter (phase B)
ILc	Phase current before LCL filter (phase C)
Ia	Phase current after LCL filter (phase A)
Ib	Phase current after LCL filter (phase B)
Ic	Phase current after LCL filter (phase C)

3 abc to dq conversion

The abc to dq transformation is used to convert time-varying, three-phase AC quantities voltages and current quantities into two constant DC components. The main purpose is to simplifies the analysis and control of three-phase inverter. Working with sinusoidal, time-varying AC signals makes PI controller operation difficult. By rotating the reference frame to align with the electrical frequency, d and q become constant DC-like values. This makes it vastly easier to apply standard control theory.

Transformation steps :

- Clarke transformation (abc to $\alpha\beta$) : Converts the stationary three-phase variables (a, b, c) into a stationary two-phase orthogonal reference frame (α, β)

  The three phase voltages : V_a , V_b , V_c

  Phase angle = θ

  For a balanced three phase system : $V_a + V_b + V_c = 0$

  Clarke transformation :

$$\begin{bmatrix} V_\alpha \\ V_\beta \end{bmatrix} = \frac{2}{3}\begin{bmatrix} 1 & -\frac{1}{2} & -\frac{1}{2} \\ 0 & \frac{\sqrt{3}}{2} & -\frac{\sqrt{3}}{2} \end{bmatrix}\begin{bmatrix} Va \\ V_b \\ V_c \end{bmatrix}$$

$$V_\alpha = \frac{2}{3}\left(V_a - \frac{V_b}{2} - \frac{V_c}{2}\right)$$

$$V_\beta = \frac{2}{3}\left(\frac{\sqrt{3}V_b}{2} - \frac{\sqrt{3}V_c}{2}\right)$$

Now the three phase quantities have become two orthogonal stationary axis quantities.

- Park transformation ($\alpha\beta$ to dq): Rotates the αβ reference frame at the same frequency as the AC system, locking it to the angle of the rotor or voltage vector. This outputs the d-axis (direct) and q-axis (quadrature) components, which behave like DC values.

Now rotating the stationary $\propto \beta$ frame by angle θ

$$\begin{bmatrix} V_d \\ V_q \end{bmatrix} = \begin{bmatrix} cos\theta & sin\theta \\ -sin\theta & cos\theta \end{bmatrix}\begin{bmatrix} V_\propto \\ V\_\beta \end{bmatrix}$$

$$V_d = V_\propto cos\theta + V_\beta sin\theta$$

$$V_q = -V_\propto sin\theta + V_\beta cos\theta$$

4 Voltage controller loop

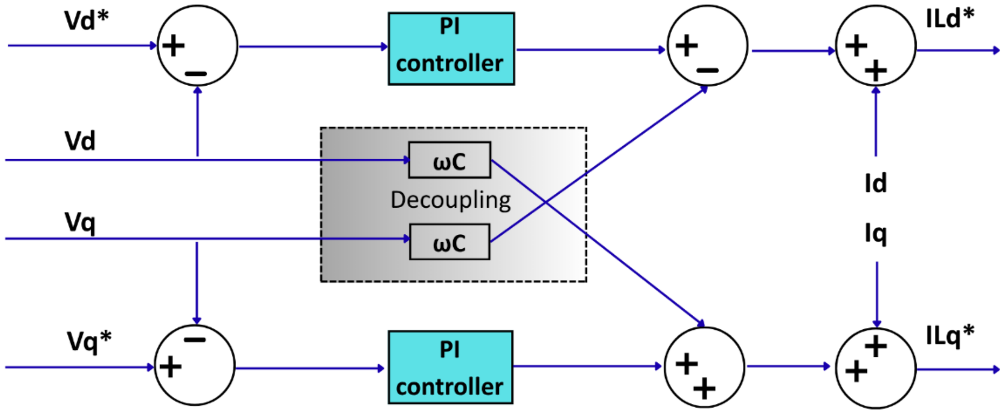

Figure 2 : Voltage controller loop

- The voltage control loop regulates the inverter output voltage by comparing the measured output voltages with their respective reference values in the synchronous *dq* reference frame.
- The measured direct-axis (Vd) and quadrature-axis (Vq) voltages are subtracted from the reference voltages(Vd* and Vq*) to generate the voltage error signals .
- Error signals are processed by a Proportional-Integral (PI)controller to minimize the steady-state voltage error and generates the intermediate control signals.
- To improve the dynamic performance of the controller, cross-coupling compensation (decoupling) is performed. The coupling terms introduced by the synchronous rotating reference frame are compensated using the feedforward signals ωC×Vq and ωC×Vd. These terms reduce the interaction between the d- and q-axes, allowing the voltage controllers to operate independently and improving the transient response. Here, C is the capacitance of the filter capacitor per phase in the LCL filter.
- The measured phase currents after LCL (Id and Iq) are feedforwarded to enhance disturbance rejection and improves the dynamic response of the voltage control loop. The outputs of the PI controllers, together with the decoupling and feedforward compensation terms, are combined to generate the reference current commands ILd* and ILq*.

5 Current controller loop

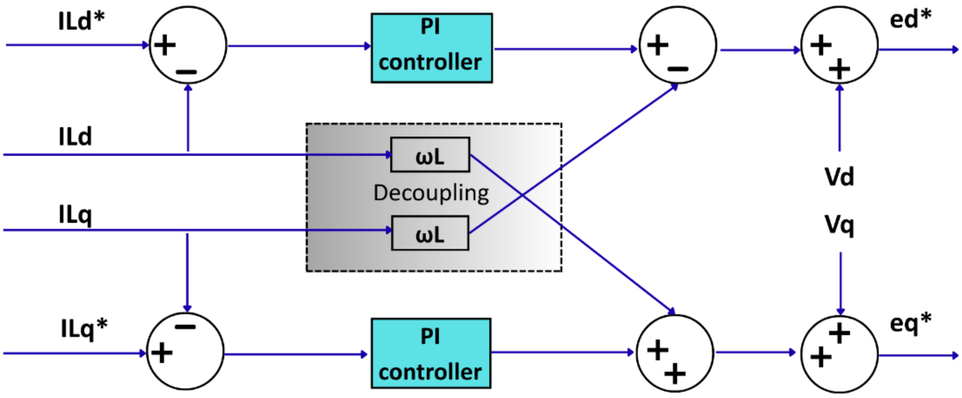

Figure 3: Current controller loop

- The current controller loop is responsible for regulating the inverter side inductor currents in the synchronous *dq* reference frame. The reference current components (ILd* and ILq*) generated by the outer voltage control loop, are compared with the measured inductor currents (Id and Iq) to produce the current error signals.
- The current error signals are processed by a Proportional-Integral (PI) controller to minimize the current tracking error by eliminating the steady state error while providing a fast dynamic response. The controller outputs represent the required control voltages needed to regulate the inverter side inductor currents.
- A decoupling network is incorporated using the terms ωL×ILd and ωL×ILq. These cross-coupling compensation terms cancel the coupling introduced by the synchronous rotating reference frame, enabling independent control of the two current components and improving the dynamic performance of the controller. (L represents the inductance of each phase of the LCL filter. Since a symmetrical LCL filter is employed in the proposed design, the inverter side and grid side inductors are selected with equal inductance values (L1 = L2).
- The measured capacitor voltages (Vd and Vq) are entered as feedforward compensation. The feedforward voltages improve the controller response by compensating for output voltage disturbances and reducing the control effort required from the PI controller.
- The outputs of the PI controllers, together with the decoupling and voltage feed-forward terms, are combined to generate the voltage reference signals (ed* and eq*)

6 dq to abc conversion

The generated voltage references are transformed from dq frame to three-phase abc reference voltages. These three-phase voltage references are then applied to the PWM generator, where they are compared with the carrier signal to generate the switching pulses for the inverter MOSFETs.

- Inverse Park Transformation :

  The d axis and q axis voltages = $e_d *, e_q *$
  Phase angle = θ
  Stationary frame voltages :

$$\begin{bmatrix} V_\propto \\ V_\beta \end{bmatrix} = \begin{bmatrix} cos\theta & -sin\theta \\ sin\theta & cos\theta \end{bmatrix} \begin{bmatrix} V_d \\ V_q \end{bmatrix}$$

$$V_\propto = V_d cos\theta - V_q sin\theta$$

$$V_\beta = V_d sin\theta + V_q cos\theta$$

- The inverse Clarke transformation

$$\begin{bmatrix} V_a \\ V_b \\ V_c \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ -\frac{1}{2} & \frac{\sqrt{3}}{2} \\ -\frac{1}{2} & -\frac{\sqrt{3}}{2} \end{bmatrix} \begin{bmatrix} V_\propto \\ V_\beta \end{bmatrix}$$

$$V_a = V_\propto$$

$$V_b = -\frac{V_\propto}{2} + \frac{\sqrt{3}V_\beta}{2}$$

$$V_c = -\frac{V_\propto}{2} - \frac{\sqrt{3}V_\beta}{2}$$

7 PWM generation

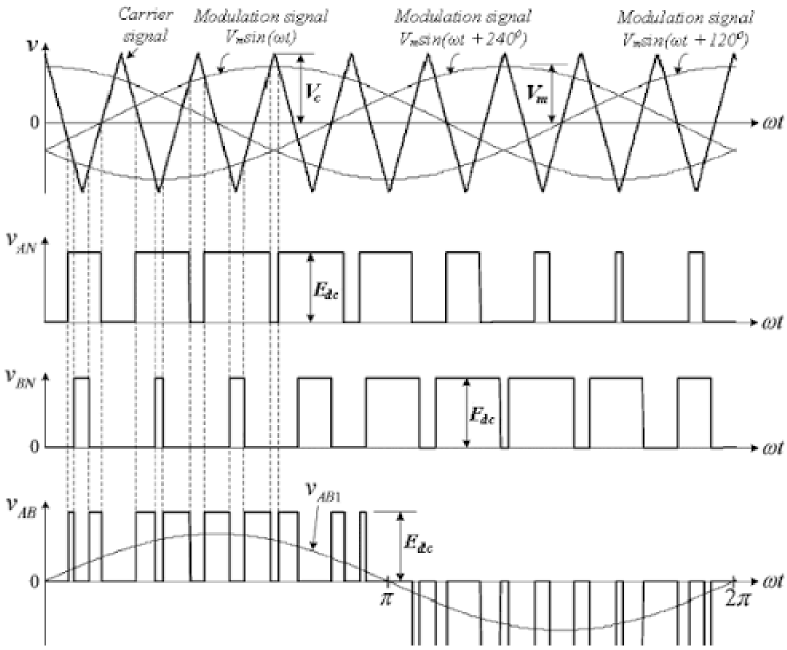

Figure 4: SPWM for PWM signal generation . Adapted from[1]

The PWM (Pulse Width Modulation) generation module is responsible for producing the gate signals required to drive the six MOSFETs in the Three-phase inverter bridge. The STM32F411 microcontroller generates three sinusoidal reference voltages, displaced by 120°, through the control algorithm. These reference voltages are compared with a 10 kHz triangular carrier waveform using the Sinusoidal Pulse Width Modulation (SPWM) technique.

The comparison process produces six complementary PWM signals, which are supplied to the IR2110 gate driver circuits. By continuously adjusting the PWM duty cycle according to the reference voltages, the inverter synthesizes a balanced three-phase AC voltage. The high frequency PWM waveform is subsequently filtered by the LCL filter to obtain a near sinusoidal output voltage suitable for supplying islanded loads.

8 Bibliography

[1] Qamil Kabashi, Myzafere Limani, Nebi Caka, and Milaim Zabeli, “Low Order Harmonic Analysis of 3-Phase SPWM and SV-PWM Inverter Systems Driving an Unbalanced 3-Phase Induction Motor Load,” May 2018.